

Kuan Wei Chen

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Education

National Central University, Taoyuan, Taiwan

Starting Sep 2025

Master in Computer Science, specialization in Artificial Intelligence

National Changhua University of Education, Changhua, Taiwan

Sep 2021 – Jun 2025

Bachelor in Electrical Engineering, GPA 3.8/4.0 | Grad. Cert.

Experience

Project Assistant in AI Application Laboratory

Jun 2025 – Dec 2025

Certificate

- Assisted in TEEP and IIPP projects, collaborating with 8 international interns.
- Contributed to “Multi-biometric Authentication Systems” and “Action Recognition Systems”.

Publications

STRLite: Lightweight Masked Autoencoders for Adaptable Scene Text Recognition

IEEE ICCE

GitHub

- Co-first author; proposed a 6M-parameter MAE plus autoregressive STR pipeline for low-cost domain adaptation.
- Achieved 93.82% weighted accuracy on six standard STR benchmarks and 81.03% on U14M dataset.

Towards Compact Sign Language Translation: Frame Rate and Model Size Trade-offs

IEEE ICCE

- First author; built a 77M-parameter MMPose-to-T5-small gloss-free SLT pipeline.
- Showed that 12 fps cuts complexity by 75% approximately with only a modest BLEU drop from 10.06 to 9.53.

Two-step Authentication: Multi-biometric System Using Voice and Facial Recognition

IET ICETA

arXiv | GitHub | Certificate

- First author; led a team of 2 international interns.
- Designed a ResNet-based voice verification architecture and utilize VGG16 for facial recognition in TensorFlow.

Projects

SignDATA: Data Pipeline for Sign Language Translation

Python, Det, Pose, System Design

arXiv | GitHub

- First author; Open-source GitHub repo reached 100+ stars; Built a config-driven preprocessing toolkit that standardizes heterogeneous sign-language datasets into training-ready artifacts.
- designed pose and video recipes covering localization, clipping, cropping, landmark extraction, and export.

Skills

Languages

English (IELTS 6.5), Mandarin (native), Hokkien (native)

Programming

Python, C/C++

Libraries/Frameworks

PyTorch, TensorFlow, Hugging Face, Scikit-learn, NumPy, Pandas

Tools/Platforms

Git, GitHub

Awards and Certifications

Micro-Program in Green Energy Technology

Electrical Engineering, NCUE

- Courses included “Introductions to Green Energy Technology,” “Circuit Theory Experiments,” “Circuit Theory,” and “Special Topic for Industry.”

Teaching Assistant – Linear Algebra

Electrical Engineering, NCUE

Certificate

- Created and delivered supplementary materials for 40+ students.
- Assisted with exam invigilation.

Model United Nations Certifications

2018–2019

HSNU MUN VII | TCGSMUN VII | Central Taiwan MUN IX | Central Taiwan MUN XII